

Multi-Process Restaurant Simulation

1. Developed a multi-process Restaurant Simulation System in C using `fork()` to create concurrent processes for customers, waiters, and cooks, ensuring parallel execution with synchronised access to shared resources.
2. Implemented Inter-Process Communication (IPC) using `shmget()`, `shmat()`, and `semget()` for shared memory and semaphores, coupled with `semop()` for atomic operations (`sem_wait/sem_signal`) to prevent race conditions.
3. Optimised timing and scheduling with `usleep()` to simulate real-world delays (e.g., food preparation, customer eating time) while maintaining synchronisation across processes.
4. Designed a priority-based queueing system using circular buffers in shared memory, ensuring fair order processing and efficient resource allocation among forked processes.

GitHub Link: <https://github.com/adhi0987/Mutli-Process-Restaurant-Simulation>

Screenshots:

```
adhithya@pop-os:~/Desktop/Mutli-Process-Restaurant-Simulation$ ./cook
[11:00 am] Cook C is ready
[11:00 am] Cook D is ready
[11:01 am] Cook C: Preparing order (Waiter U, Customer 1, Count 4)
[11:01 am] Cook D: Preparing order (Waiter V, Customer 2, Count 1)
[11:06 am] Cook D: Prepared order (Waiter V, Customer 2, Count 1)
[11:06 am] Cook D: Preparing order (Waiter W, Customer 3, Count 1)
[11:11 am] Cook D: Prepared order (Waiter W, Customer 3, Count 1)
[11:11 am] Cook D: Preparing order (Waiter X, Customer 4, Count 1)
[11:16 am] Cook D: Prepared order (Waiter X, Customer 4, Count 1)
[11:16 am] Cook D: Preparing order (Waiter Y, Customer 5, Count 3)
[11:21 am] Cook C: Prepared order (Waiter U, Customer 1, Count 4)
[11:21 am] Cook C: Preparing order (Waiter V, Customer 7, Count 2)
[11:31 am] Cook C: Prepared order (Waiter V, Customer 7, Count 2)
[11:31 am] Cook C: Preparing order (Waiter U, Customer 6, Count 2)
[11:31 am] Cook D: Prepared order (Waiter Y, Customer 5, Count 3)
[11:31 am] Cook D: Preparing order (Waiter W, Customer 8, Count 1)
[11:36 am] Cook D: Prepared order (Waiter W, Customer 8, Count 1)
[11:36 am] Cook D: Preparing order (Waiter X, Customer 9, Count 2)
[11:41 am] Cook C: Prepared order (Waiter U, Customer 6, Count 2)
[11:41 am] Cook C: Preparing order (Waiter Y, Customer 10, Count 3)
[11:46 am] Cook D: Prepared order (Waiter X, Customer 9, Count 2)
[11:46 am] Cook D: Preparing order (Waiter U, Customer 12, Count 1)
```

```

adithya@pop-os:~/Desktop/Mutli-Process-Restaurant-Simulation$ ./waiter
[11:00 am] Waiter U is ready
[11:00 am]     Waiter V is ready
[11:00 am]         Waiter W is ready
[11:00 am]             Waiter X is ready
[11:00 am]                 Waiter Y is ready
[11:01 am]     Waiter V: Placing order for Customer 2 (count = 1)
[11:01 am] Waiter U: Placing order for Customer 1 (count = 4)
[11:01 am]     Waiter W: Placing order for Customer 3 (count = 1)
[11:01 am]         Waiter X: Placing order for Customer 4 (count = 1)
[11:01 am]             Waiter Y: Placing order for Customer 5 (count = 3)
[11:02 am]     Waiter V: Placing order for Customer 7 (count = 2)
[11:02 am] Waiter U: Placing order for Customer 6 (count = 2)
[11:06 am]     Waiter V: Serving food to Customer 2
[11:09 am]         Waiter W: Placing order for Customer 8 (count = 1)
[11:11 am]             Waiter W: Serving food to Customer 3
[11:16 am]                 Waiter X: Serving food to Customer 4
[11:18 am]                     Waiter X: Placing order for Customer 9 (count = 2)
[11:21 am] Waiter U: Serving food to Customer 1
[11:25 am]             Waiter Y: Placing order for Customer 10 (count = 3)
[11:31 am]         Waiter V: Serving food to Customer 7
[11:31 am]             Waiter Y: Serving food to Customer 5
[11:36 am]                 Waiter W: Serving food to Customer 8

```

```

adithya@pop-os:~/Desktop/Mutli-Process-Restaurant-Simulation$ ./customer
[11:00 am] Customer 1 arrives (count = 4)
[11:00 am] Customer 2 arrives (count = 1)
[11:00 am] Customer 3 arrives (count = 1)
[11:00 am] Customer 4 arrives (count = 1)
[11:00 am] Customer 5 arrives (count = 3)
[11:00 am] Customer 6 arrives (count = 2)
[11:00 am] Customer 7 arrives (count = 2)
[11:01 am]     Customer 2: Order placed to Waiter V
[11:01 am]     Customer 1: Order placed to Waiter U
[11:01 am]     Customer 3: Order placed to Waiter W
[11:01 am]     Customer 4: Order placed to Waiter X
[11:01 am]     Customer 5: Order placed to Waiter Y
[11:02 am]     Customer 7: Order placed to Waiter V
[11:02 am]     Customer 6: Order placed to Waiter U
[11:06 am]         Customer 2 gets food [Waiting time = 6]
[11:08 am] Customer 8 arrives (count = 1)
[11:09 am]     Customer 8: Order placed to Waiter W
[11:11 am]         Customer 3 gets food [Waiting time = 11]
[11:16 am]         Customer 4 gets food [Waiting time = 16]
[11:17 am] Customer 9 arrives (count = 2)
[11:18 am]     Customer 9: Order placed to Waiter X
[11:21 am]         Customer 1 gets food [Waiting time = 21]

```